

CBAM and CoordAtt in YOLOv8n: Attention Location Effects on KITTI Occlusion Detection

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Abstract—Object detection using deep learning, particularly with occluded objects, remains challenging. The YOLOv8n architecture, popular for real-time detection, can be enhanced with attention modules. This research evaluates the effect of the insertion location (Neck, Backbone, or Both) of two attention modules, CBAM (Convolutional Block Attention Module) and CoordAtt, on YOLOv8n's performance in detecting occluded objects. Experiments were conducted on a modified KITTI dataset with explicit occlusion level labels for six combined vehicle and pedestrian classes. Results consistently demonstrate that inserting attention modules into the Neck section yields the most significant performance improvement over the baseline and other locations. Specifically, the CBAM-Neck configuration achieved the highest mAP@0.5 (0.683) and mAP@0.5:0.95 (0.476), a substantial improvement from the baseline (0.659 and 0.449). Qualitative analysis confirmed enhanced capability in detecting moderately occluded objects, although severe occlusion remains a challenge. Backbone or Both insertions proved less effective.

Keywords—YOLOv8, CBAM, CoordAtt, object detection, occlusion, attention, KITTI

I. INTRODUCTION

Object detection stands as a fundamental task in computer vision, underpinning critical applications such as autonomous driving systems and intelligent surveillance [1], [2]. The advent of deep learning, especially **Convolutional Neural Networks (CNNs)**, has propelled significant advancements in detection accuracy and robustness [1]. Among various architectures, the **YOLO (You Only Look Once)** family [3], including its recent iteration YOLOv8n [4], has gained prominence due to its effective balance between processing speed (real-time capability) and detection accuracy.

Despite these advances, performance often suffers in complex real-world scenarios, most notably when target objects are subject to occlusion—being partially or fully obscured by other objects [5], [6]. Occlusion leads to the loss of crucial visual information, such as shape details and texture, drastically reducing detector accuracy [7], [8]. This performance degradation is a critical issue for safety-sensitive applications like autonomous navigation, where failing to detect an occluded pedestrian or vehicle can have severe consequences [2]. The challenge of occlusion can be categorized into several types, including inter-class occlusion (e.g., a car obscuring a pedestrian), intra-class occlusion (e.g., overlapping pedestrians in a crowd), and self-occlusion (e.g., unusual object poses). Each type presents a unique challenge to

a detector's ability to infer the complete object shape and class from partial evidence.

Attention mechanisms [9], inspired by the human visual system's ability to selectively focus on relevant information, have emerged as a promising strategy to mitigate these challenges within CNNs [10], [11]. By enabling models to adaptively weight and emphasize salient features, attention mechanisms can potentially enhance discriminative power, even when parts of an object are not visible [10]. Two widely recognized and effective attention modules are **CBAM (Convolutional Block Attention Module)** [10], which sequentially combines channel-wise and spatial attention, and **CoordAtt (Coordinate Attention)** [11], which captures precise positional information. These two were chosen as they are lightweight and represent different successful strategies (sequential refinement vs. coordinate embedding), making their comparison insightful.

While integrating attention holds promise, the question of where to optimally insert these modules remains largely unanswered. Does incorporating attention early in the feature extraction Backbone provide more benefit than applying it later during the multi-scale feature fusion process in the Neck [12], [13]? Different insertion points could significantly impact performance [14], [15]. This paper addresses these questions through a systematic investigation on a modified **KITTI (Karlsruhe Institute of Technology and Toyota Technological Institute)** dataset [16]. Our primary contribution lies in providing empirical evidence on the interplay between attention module type, insertion location, and occlusion detection performance within the YOLOv8n framework, offering clearer architectural design principles for real-world conditions.

II. RELATED WORK

A. YOLO Architectures

The YOLO family [3] marked a significant shift towards real-time object detection with its single-stage methodology. A typical YOLOv8 structure (Fig. 1) comprises three main components: a **Backbone** for hierarchical feature extraction; a **Neck**, which employs mechanisms like **Feature Pyramid Network (FPN)** [12] and **Path Aggregation Network (PANet)** [13] to fuse features from different levels; and a **Head**, which generates the final predictions [4], [17].

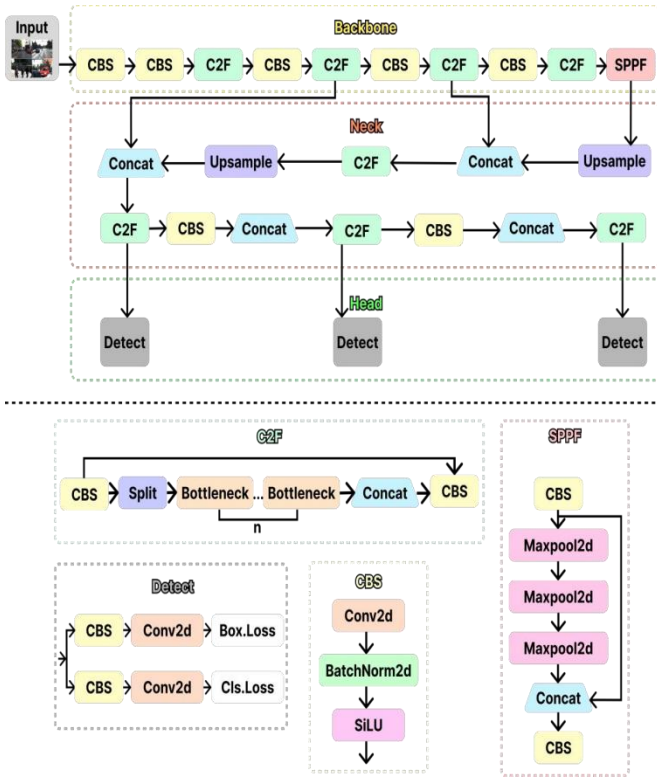
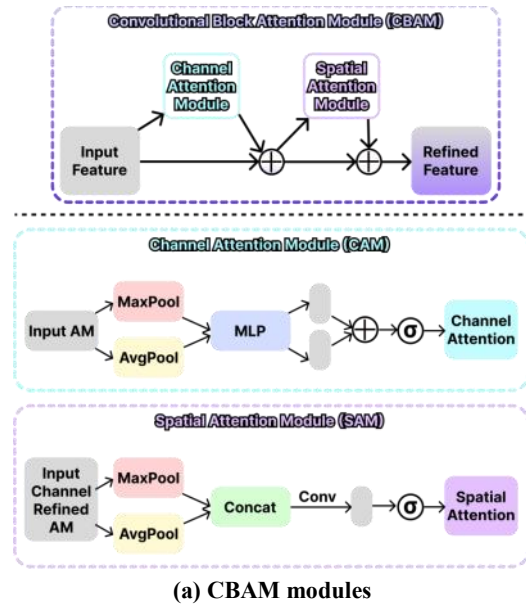


Fig 1. General architecture of the YOLOv8 model, showing the Backbone, Neck (with FPN and PANet), and decoupled Head components.

The flexibility of YOLOv8's modular design, particularly its configuration via YAML files, allows for straightforward modification, such as the insertion of custom layers like attention modules. The Backbone, typically based on CSPDarknet principles, is responsible for creating a rich feature hierarchy. The Neck's role is crucial for detecting objects at various scales by effectively combining low-level features (with high spatial resolution) and high-level features (with rich semantic information). Finally, the anchor-free, decoupled Head separates the classification and regression tasks, often leading to improved performance.

B. Attention Mechanisms

Attention mechanisms enhance neural networks by enabling them to focus on informative features [9]. CBAM [10] does this sequentially. First, a channel attention module determines "what" to focus on by learning weights for each feature channel. Then, a spatial attention module determines "where" by creating a 2D attention map. Its conceptual workflow is shown in Fig. 2(a). CoordAtt [11], in contrast, explicitly encodes coordinate information to capture long-range dependencies. It achieves this by decomposing the channel attention into two parallel 1D feature encoding steps, one for the horizontal axis and one for the vertical axis, as shown in Fig. 2(b). This approach addresses a known limitation in simpler attention mechanisms like Squeeze-and-Excitation (SE) [21], which rely on global pooling and lose positional information [11].



(a) CBAM modules

(b) CoordAtt modules

Fig 2. Simplified workflows of (a) CBAM and (b) CoordAtt modules.

C. Occlusion Handling and Attention in YOLO

Occlusion is a significant and persistent challenge in object detection [5]. Attention mechanisms offer a data-driven approach to adaptively focus on visible object parts or surrounding context to infer occluded parts [6], [9]. Several studies have integrated attention into YOLO architectures. For instance, some research reported improvements in detecting dense, overlapping pedestrians by adding CBAM to YOLOv8 [14] or improving general vehicle-pedestrian detection [20]. Others have specifically targeted occluded vehicles [7] or used feature fusion alongside CBAM for occluded pedestrians [6]. However, these works often focus on a single insertion strategy or do not systematically compare different attention modules. While Yang et al. [6] used CBAM with a multiscale approach and He et al. [7] proposed a specialized attention for occluded vehicles, our study provides a direct, systematic comparison of two popular, general-purpose attention modules (CBAM vs. CoordAtt) across three distinct architectural locations (Neck vs. Backbone vs. Both). This controlled comparison on a dataset with explicit occlusion labels is a gap in existing literature, and our work aims to provide clearer design principles for attention placement in YOLOv8.

III. PROPOSED METHOD

Our methodology focuses on modifying the standard YOLOv8n [4] architecture by systematically integrating CBAM [10] and CoordAtt [11] at various strategic locations to evaluate their impact on occlusion detection.

A. Architecture Modification

We created six modified YOLOv8n architectures, in addition to the baseline model, resulting in seven models for comparison. The six variants are: CBAM-Neck, CBAM-Backbone, CBAM-Both, CoordAtt-Neck, CoordAtt-Backbone, and CoordAtt-Both. The insertion points were strategically chosen to be immediately after specific C2f blocks [17], which are fundamental building blocks in YOLOv8. The three targeted architectural sections were:

- **Neck:** Modules inserted after C2f blocks within the FPN/PANet feature fusion layers (at original indices 15, 18, and 21). This targets the refinement of fused, multi-scale semantic features.
- **Backbone:** Modules inserted after C2f blocks within the initial feature extraction pathway (at original indices 4, 6, and 8). This tests the effect of refining features at early hierarchical levels.
- **Both:** Modules inserted at all six specified locations in the Neck and Backbone simultaneously, to investigate a more aggressive attention strategy.

Figure 3 provides a conceptual visualization of where these attention blocks could be added within the YOLOv8n pipeline.

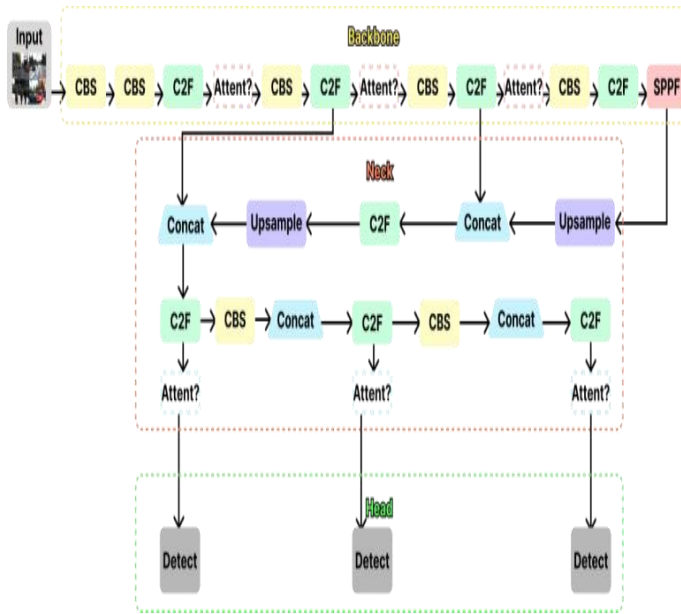


Fig 3. Conceptual diagram illustrating potential insertion points (?) for attention modules within the YOLOv8n Neck and Backbone.

The precise layer indices and corresponding channel dimensions for each insertion point are detailed in Table 1, which is critical for correct module initialization.

Table 1. Attention Module Insertion Points and Input Channel

Location	Index	Layer Type	Channels
Backbone 1	4	C2f	64
Backbone 2	6	C2f	128
Backbone 3	8	C2f	256
Neck 1	15	C2f	64
Neck 2	18	C2f	128
Neck 3	21	C2f	256

The choice of placing attention modules after C2f blocks is deliberate. C2f blocks are fundamental components in YOLOv8 that process and refine features. Inserting attention immediately after these blocks allows the model to re-weight and re-calibrate the feature maps produced by the C2f blocks before they are passed to the next stage, potentially enhancing the most salient information for detection.

B. Implementation Details

The integration of attention modules was achieved by programmatically modifying the standard YOLOv8n YAML (YAML Ain't Markup Language) configuration files. We first inserted a placeholder `torch.nn.Identity` layer at each designated location. This strategy allowed us to leverage the standard model loading capabilities of the Ultralytics framework to seamlessly transfer weights from a pre-trained baseline YOLOv8n model to all the unmodified layers of our custom architectures. Following the initial weight loading, a custom Python script programmatically iterated through the layers of the loaded model. Upon encountering an `nn.Identity` placeholder, the script replaced it with an actual instance of either the CBAM or CoordAtt module, initialized with the correct number of input/output channels as specified in Table 1. This ensured architectural compatibility and allowed for effective fine-tuning.

C. Dataset and Preprocessing

The KITTI dataset [16] formed the foundation for our experiments. To enable a more targeted analysis of occlusion handling, we performed two specific modifications:

- **Class Consolidation:** Similar object classes were merged. 'Car', 'Van', and 'Truck' were consolidated into `kendaraan`. 'Pedestrian' became `pejalan_kaki`.
- **Explicit Occlusion Labeling:** The original KITTI occlusion levels (0: fully visible, 1: partly occluded, 2: largely occluded/difficult) were integrated into the class labels using numerical suffixes (`_00`, `_01`, `_11`). This resulted in 6 distinct target classes (e.g., `kendaraan_00`, `kendaraan_01`), allowing the model evaluation to explicitly differentiate performance based on occlusion severity.

All input images were uniformly resized to 640x640 pixels. Standard data augmentation techniques from the Ultralytics library, such as mosaic augmentation, were applied during training to enhance model robustness.

D. Training Protocol

A consistent and controlled two-phase training protocol was adopted for all experiments to ensure fair comparisons:

- **Baseline Training:** The standard YOLOv8n model was trained from scratch on our modified KITTI training dataset for 100 epochs.
- **Fine-tuning:** Each of the six attention-modified architectures was subsequently fine-tuned for an additional 100 epochs, initialized using the best weights from the baseline training. This transfer learning approach focuses the training on adapting the network to the integrated attention modules.

Specific hyperparameters were set for fine-tuning: an initial learning rate of 0.0001 and an early stopping patience of 50 epochs. The validation loss was closely monitored against the training loss to ensure that no significant overfitting occurred. A consistent batch size of 16 was used across all runs. Training efficiency was enhanced by utilizing **Automatic Mixed Precision (AMP)**. All experiments were performed on the Kaggle Notebooks platform, utilizing NVIDIA Tesla P100 GPUs.

E. Evaluation Metrics

To comprehensively assess model performance, we employed a suite of standard object detection metrics on the validation set:

- **mAP@0.5:** Mean Average Precision at an IoU threshold of 0.5 [18].
- **mAP@0.5:0.95:** mAP averaged over IoU thresholds from 0.5 to 0.95 [19], providing a stricter assessment of localization accuracy.
- **Precision and Recall [2]:** These metrics measure the rate of correct positive detections and the rate of actual objects found, respectively.

Qualitative evaluation included visual inspection of predictions and analysis of confusion matrices.

IV. EXPERIMENTS AND RESULTS

A. Quantitative Results

The primary quantitative outcomes comparing the baseline model with the six attention-augmented variants are summarized in Table 2.

Table 2. Quantitative Performance Comparison

Model Configuration	mAP @.5	mAP @.5:.95	Precision	Recall
YOLOv8n baseline	0.659	0.449	0.679	0.629
CBAM-Neck	0.683	0.476	0.751	0.620
CBAM-Backbone	0.671	0.457	0.686	0.624
CBAM-Both	0.671	0.457	0.686	0.624

CoordAtt-Neck	0.682	0.469	0.675	0.678
CoordAtt-Backbone	0.671	0.457	0.686	0.624
CoordAtt-Both	0.671	0.457	0.686	0.624

The results in Table 2 consistently highlight the superior effectiveness of inserting attention modules within the **Neck**. Both CBAM-Neck and CoordAtt-Neck demonstrate significant improvements in mAP scores. Notably, CBAM-Neck achieved the highest performance in both mAP@0.5 (0.683) and mAP@0.5:0.95 (0.476), representing substantial gains of **2.4%** and **2.7%**, respectively, over the baseline.

In stark contrast, inserting attention modules solely within the Backbone or in Both locations provided minimal or no discernible performance advantage. This strongly supports the conclusion that the Neck, where multi-scale features are fused, is the most advantageous location for attention module integration.

When comparing the two attention mechanisms at this optimal Neck location, a clear trade-off emerges. CBAM achieves higher **Precision** (0.751), indicating fewer false positives. Conversely, CoordAtt achieves a much higher **Recall** (0.678), suggesting it is better at identifying relevant objects, even if they are difficult to detect. This trade-off has significant practical implications. In autonomous driving, failing to detect a pedestrian (false negative) is often more catastrophic than incorrectly detecting a non-existent one (false positive). Therefore, the higher recall of CoordAtt-Neck might be preferable in safety-critical applications. To gain deeper insights beyond these quantitative metrics, we performed a qualitative analysis.

B. Qualitative Analysis

Visual inspection of detection results on sample images from the validation set provides deeper insights and supports the quantitative findings. Figure 4 presents an example comparing the detections of different models on a complex image containing several vehicles, some under partial occlusion.



Fig 4. Visual detection results on a sample image (e.g., 004963) showing Ground Truth (top-left), Baseline YOLOv8n (top-right), CBAM-Neck (bottom-left), and CoordAtt-Neck (bottom-right).

In this example, the baseline YOLOv8n (top-right panel of Figure 4) fails to detect a partially occluded silver/white vehicle (labeled "Kendaraan (11)" in the Ground Truth) located in the background on the far right. Additionally, it also misses a pedestrian (labeled "Pejalan kaki (00)" in the

Ground Truth) positioned partially obscured behind another car on the right side of the road. In stark contrast, both the CBAM-Neck (bottom-left panel of Figure 4) and CoordAtt-Neck (bottom-right panel of Figure 4) successfully identify these challenging objects, with visible bounding boxes and corresponding labels.

This visual evidence corroborates the quantitative results, showing that attention integration within the Neck section enhances the model's sensitivity to partial visual cues. The bounding boxes generated by the Neck-modified architectures also appear more accurate and tightly fitted to the visible portions of the objects compared to the baseline.

V. CONCLUSION

This paper conducted a systematic evaluation of the influence of attention module type (CBAM, CoordAtt) and insertion location (Neck, Backbone, Both) on the occlusion handling capability of the YOLOv8n detector. Our investigation yields several key conclusions. First, the insertion location of attention modules is of paramount importance. Integrating attention within the Neck section consistently provided significantly superior performance improvements compared to insertions within the Backbone or Both locations. This suggests that refining fused, multi-scale features just before the detection head is the most effective strategy. Second, both CBAM and CoordAtt proved to be effective mechanisms for enhancing detection performance when strategically placed in the Neck. A practical trade-off was observed: CBAM-Neck offered the highest overall accuracy (mAP) and precision, making it ideal for applications where minimizing false alarms is critical. In contrast, CoordAtt-Neck excelled in recall, suggesting its strength in identifying more challenging, partially occluded objects, a crucial trait for safety-critical systems like autonomous driving. Third, the CBAM-Neck configuration emerged as the most effective architecture in terms of standard mAP metrics. Delivering notable performance gains of approximately 2.4% in mAP@0.5 and 2.7% in mAP@0.5:0.95 relative to the standard YOLOv8n baseline, it represents a robust choice for general-purpose detection enhancement. Finally, while attention integration in the Neck substantially improves the detection of partially occluded objects, the task of accurately detecting severely occluded objects remains a significant challenge. This indicates inherent limitations in handling extreme visual information loss, even with the applied attention enhancements.

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